DOMAIN-INFORMED COMPOSITE KERNEL GAUSSIAN PROCESS CLASSIFICATION FOR MOLECULAR TOXICITY PREDICTION

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MOTIVATION

Computational toxicity screening needs models that are accurate, chemically meaningful, and able to ex-press prediction uncertainty. Most toxicity models either rely on a single molecular representation or provide limited probabilistic interpretation [3].

Core idea: combine structural and physicochemical molecular similarity inside a Gaussian Process Classifier (GPC), using a domain-informed composite kernel rather than a single generic kernel.

DATA & EVALUATION

- **Dataset** : 13,949 unique molecules curated from public toxicity resources.
- **Inputs** : SMILES string with binary toxicity labels.
- **Split** : Scaffold-based evaluation to reduce structural leakage and test out-of-scaffold generalization.
- **Training Gap** : 8,000 training molecules for tractable exact gaussian process classifiers fitting.
- **Held-out test set** : 837 molecules.
- **Evaluation** : AUC, accuracy, F1-score, Brier score, kernel ablation and classifier baseline comparison

FEATURE REPRESENTATION

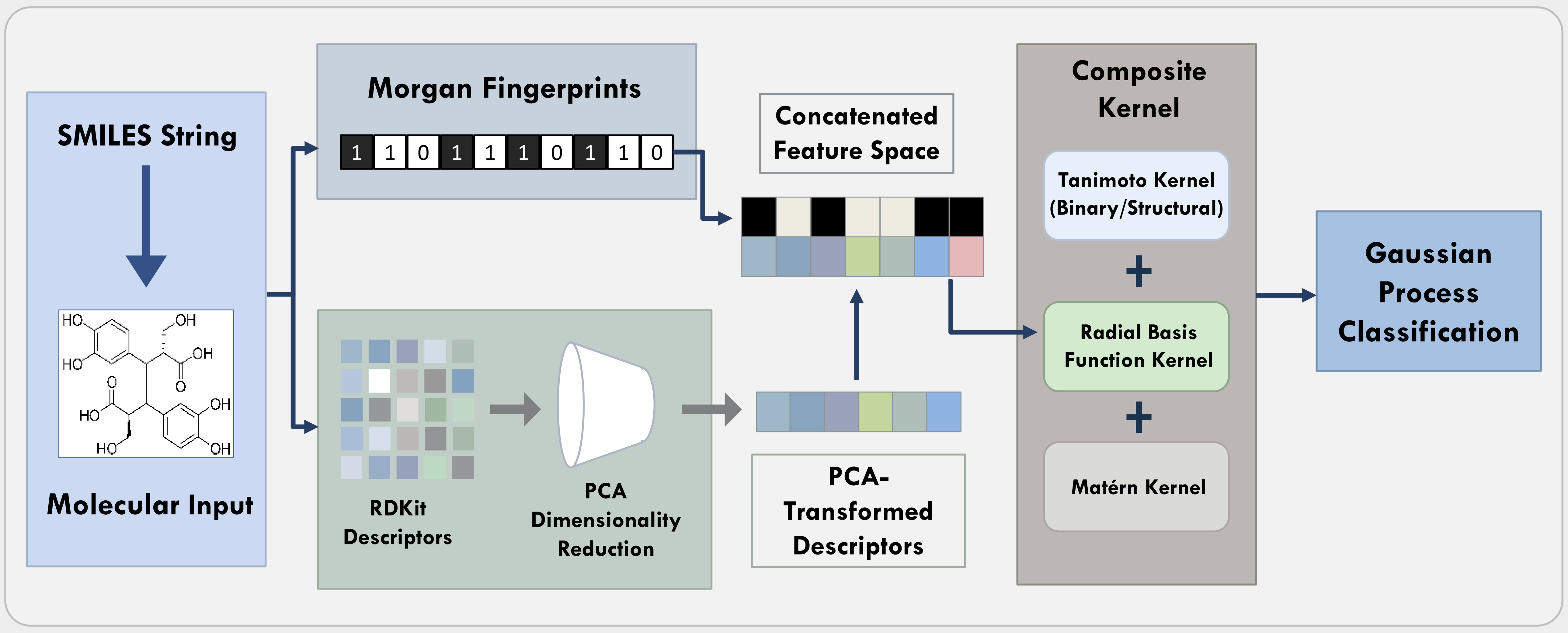
Each molecule is mapped into two complementary view:

- **Morgan fingerprints** : circular substructure fingerprints for binary structural similarity [2].
- **RDKit descriptors** : continuous physicochemical descriptors.
- **PCA descriptors view** : descriptors are stan-dardized and compressed to 128 components.
- **Composite input** : fingerprint and PCA descriptor vectors are concatenated before kernel evaluation.

INTERPRETATION

- **Representation complementarity**: structural fingerprints and physicochemical descriptors provide nonredundant toxicity signals.
- **Kernel design matters**: a domain-informed composite kernel is stronger than using Tanimoto, RBF or Matérn components in isolation.
- **Calibration matters**: the Brier score advantage indicates better probability quality, not only better ranking.

HYBRID GPC ARCHITECTURE



The architecture transforms SMILES to Morgan fingerprints and PCA-reduced RDKit descriptors, then passes the concatenated feature space through a Tanimoto + RBF + Matérn composite kernel.

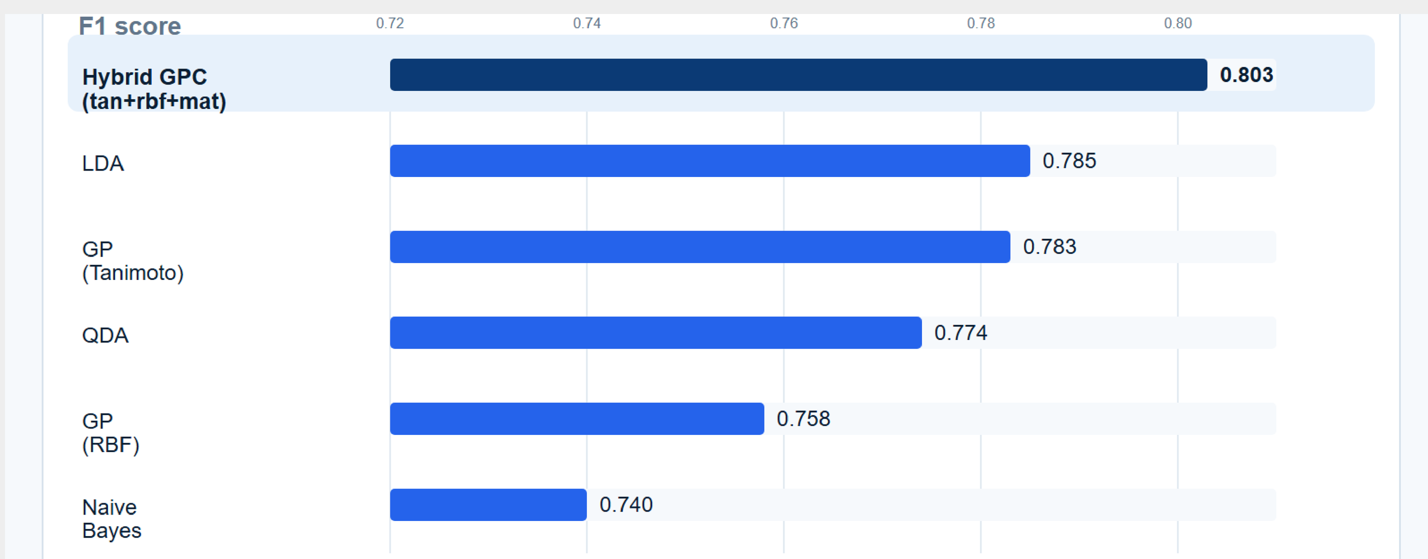
COMPOSITE KERNEL

The proposed kernel combines structural overlap and descriptor-space smoothness :

$$K(x,x') = c[w_{tan}K_{tan} + w_{rbf}K_{rbf} + w_{mat}K_{mat}] + \sigma^2 I$$

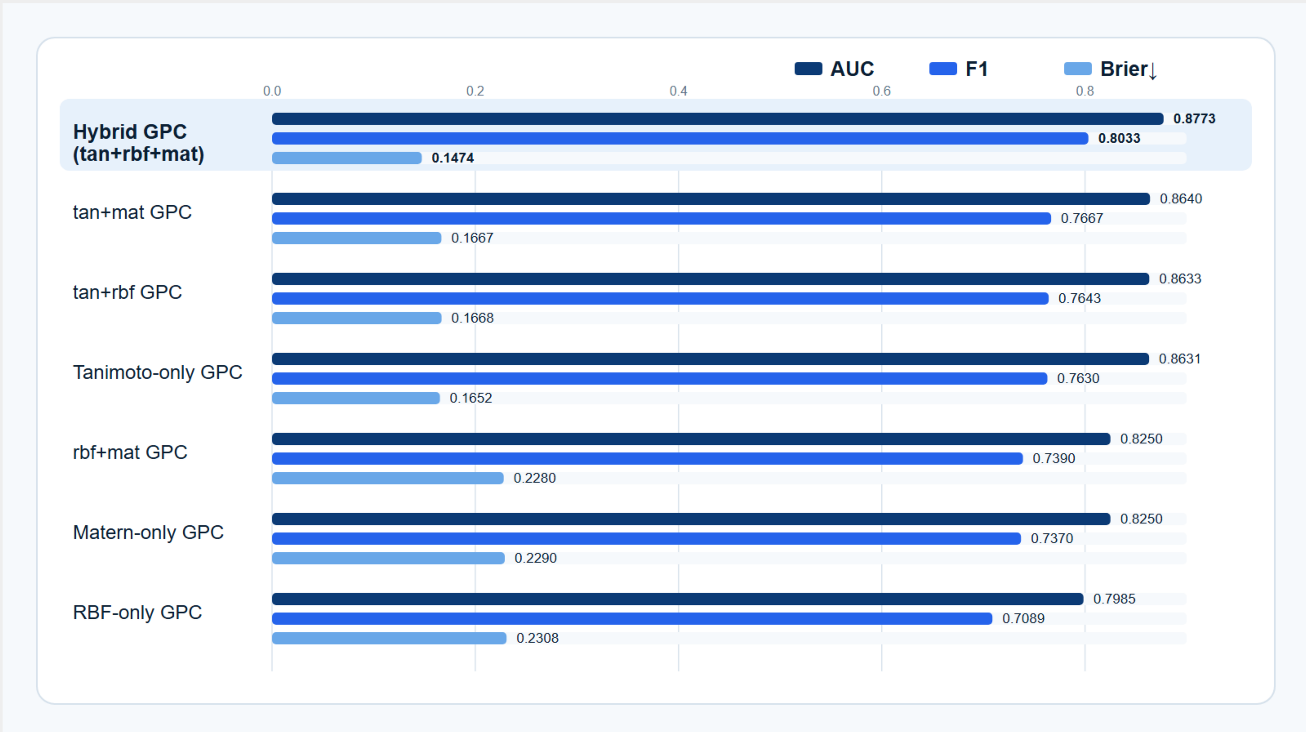
- **Tanimoto Kernel** : binary fingerprints overlap.
- **RBF Kernel** : smooth similarity over physicochemical descriptor space.
- **Matérn Kernel** : less-smooth continuous similarity, improving robustness to descriptor variation
- **GPC** : Laplace-approximation probabilistic classifier with native uncertainty estimates [1].

BASELINE COMPARISON



The Hybrid GP improves F1 against classical discriminant models and single-kernel GP baselines, while retaining probabilistic outputs needed for uncertainty-aware screening

MAIN RESULTS – KERNEL ABLATION



- The full tan + rbf + mat composite kernel gives the best overall F1/Brier/AUC trade-off, combining strong classification performance with the lowest Brier score.
- Compared with the strong Tanimoto-only GPC base-line, adding RBF and Matérn descriptor kernels improves F1 and reduces the Brier score.
- The ablation shows that structural fingerprints are essential, while descriptor-space kernels provide complementary physicochemical information for discrimination and calibration.

CONCLUSTION

- The submitted paper presents a Hybrid GPC for molecular toxicity prediction.
- The model integrates Morgan fingerprints and RD-Kit descriptors through a Tanimoto + RBF + Matérn kernel.
- The full composite kernel achieves **AUC 0.8773**, **F1-score 0.8033**, and **Brier 0.1474**
- This provides a transparent, uncertainty-aware alternative to conventional toxicity classifiers.

REFERENCES

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- [2] Álmos Orosz et al. "Comparison of Descriptor- and Fingerprint Sets in Machine Learning Models for ADME-Tox Targets". In: Frontiers in Chemistry 10 (2022), p. 852893. DOI: 10.3389/fchem.2022.852893.
- [3] Srijit Seal et al. "Machine Learning for Toxicity Prediction Using Chemical Structures: Pillars for Success in the Real World". In: Chemical Research in Toxicology 38.5 (2025), pp. 759–807. DOI: 10.1021/acs.chemrestox.5c00033